

LumoTree: Path-Parallel Speculative Verification for Hybrid Language Models

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Abstract—Tree speculative decoding for hybrid language models must preserve one coherent continuation across recurrent, convolution, and attention state. We present LumoTree, a verifier that executes recurrent paths in parallel, reuses state tiles within each path, and coordinates native recurrent replay, convolution-history gathering, and attention-cache remapping through a shared logical tree. Fused candidate selection, GPU-resident acceptance, and grouped split-K attention support the verification cycle. Component experiments show exact candidate-selection parity and recurrent agreement within paired error bounds. On a single NVIDIA DGX Spark (GB10), sampled replay of 43 recorded coding-agent requests with Qwen3.8-27B NVFP4 yields 28.79 pooled tokens/s for LumoTree and 26.12 for five-token native multi-token prediction on the same vLLM stack, over three runs each. A separate ten-task SWE-bench Verified deployment characterizes coding-agent behavior. Complete-model continuation and controlled task-completion speedups remain open.

Index Terms—speculative decoding, Gated DeltaNet, state commitment, numerical reproducibility, language-model serving

I. INTRODUCTION

Speculative decoding reduces serial target-model invocations by verifying draft tokens in parallel. Its sampling guarantee assumes that verification supplies the intended target probabilities. Tree speculation expands the candidate set, but it does not relax that assumption [1]–[3]. For recurrent-hybrid models, realizing the verifier requires more than an attention mask: a candidate must inherit the state of its ancestors, and the next serving iteration must inherit exactly the path selected for continuation [4], [5].

LumoTree executes recurrent paths in parallel, reuses state tiles across updates within each path, and coordinates the accepted continuation across recurrent, convolution, and attention state. An ancestry-aware attention path and a device acceptance walk use the same logical tree as recurrent verification and state publication. A shared descriptor carries ancestry and logical-to-physical mappings through each stage. This system boundary matters because a verifier can compute plausible candidate logits yet publish the wrong state for the next iteration.

The central execution choice is to verify branches in temporary state and replay only the selected path into the native running state. The verifier scans branch paths, while the fixed-shape committer replays accepted operands through native GDN updates in a captured graph. Their finite-precision agreement is a separate requirement, not assumed from the recurrence. This choice exposes concrete optimization questions: how much branch state to cache, when to recompute ancestry, how

to organize independent GPU work, and how to keep full-vocabulary probability arithmetic on the device. The attention layout and accepted-state mappings constrain those changes as much as the recurrent equation does.

The paper studies three coupled design questions in recurrent-hybrid tree verification. *First*, how can branch-local computation produce one coherent continuation across recurrent, convolution, and attention state? We define a shared logical tree and coordinate state publication through its logical-to-physical mapping. *Second*, how should verification trade temporary state storage against repeated computation? We use path-parallel scans with shared boundary states and replay the accepted path into persistent state, building on established chain-scheduling and replay mechanisms [6], [7]. *Third*, how can this organization reduce serving overhead? We combine fused candidate selection, device-resident acceptance, and grouped split-K attention with a spine-contiguous KV layout. We evaluate component numerical agreement, recorded-request replay against native decoding, and an exploratory coding-agent deployment. Section IX bounds these results.

The target workload is a long-running coding agent: generation alternates with repository inspection, tool execution, and tests, and a task can require many model requests. Performance therefore depends on useful completed work as well as token production. Our workload evaluation uses explicitly identified Astropy tasks from SWE-bench Verified [8]. We report the model, serving route, task cohort, and measurement definition together; a decode-rate change is not a task-completion speedup. Candidate-selection and attention tests complement the agent-workload measurements.

Tree verification and compact accepted-state reconstruction have direct prior art. Section II positions our implementation against those mechanisms. Sections III–IV develop the architecture and kernel design; Section V describes the optimizations that connect them to the serving cycle. Appendix A gives configuration and measurement details.

II. BACKGROUND AND RELATED WORK

A. Drafting, verification, and serving

Speculative sampling constructs an exact target sample from proposal and residual distributions, conditional on the probabilities actually supplied [1], [2]. SpecInfer supplies the tree-verification framing; Sequoia, EAGLE-2, and OPT-Tree investigate tree construction and its performance consequences [3], [9]–[11]. EAGLE-3 and DFlash further change the drafter, through feature-conditioned autoregression and



Fig. 1. Tree verification and accepted-path commitment. (a) Drafting provides a candidate tree and proposal probabilities q . (b) Branch-local computation produces target probabilities p and retains temporary state. (c) Sampling selects a materialized path (here a, d); recurrent replay, convolution gathering, and KV remapping publish $S(mad)$. The newly sampled token b^* remains pending. One descriptor keeps logical ancestry and physical storage aligned.

block-parallel diffusion proposals respectively [12], [13]. These drafting mechanisms are distinct from the recurrent verifier and state-publication choices studied here.

PagedAttention and SGLang address the scheduling, memory, and execution substrate [14], [15]. Marconi addresses prefix-cache admission and eviction for hybrid models [16]. Hybrid serving additionally manages recurrent-state pools alongside KV storage, as documented by SGLang’s hybrid-model integration [17], [18]. Thus neither a fast isolated recurrent kernel nor low transient state memory alone determines user-visible latency. Flash-Decoding splits the KV sequence and combines partial attention results; FlashInfer develops customizable attention with head/query fusion and split-KV execution; DeFT specializes KV grouping and partitioning to trees [19]–[21]. FastTree groups queries with shared contexts to reuse KV tiles and partitions tree work to control execution cost [22]. These attention methods establish precedents for on-chip reuse and tree-aware partitioning. Our modified FlashAttention-2 kernel combines ancestry-aware decode, grouped split-K execution, and a physical layout coordinated with accepted-path publication [23].

B. Recurrent and hybrid speculation

Gated DeltaNet augments a decayed recurrent state with a data-dependent delta update [4]. Parallel delta-rule formulations provide relevant algebraic background [24]. Snakes and Ladders introduces activation replay for SSM speculative decoding, and STree also uses replay to recover continuation state [5], [25]. Stateful speculation is established territory: STree studies speculative trees for hybrid state-space models, SpecMamba targets an FPGA implementation, and component-aware self-speculation studies how hybrid components affect the design [5], [26], [27]. These works motivate an explicit state lineage; they do not justify treating every recurrent update or numerical implementation as interchangeable.

C. Direct GDN comparisons

Trees from Marginals (Weaver) combines a factorized drafter with an autoregressive adapter. Its GDN verifier uses an ancestor-masked triangular solve and replays the selected path through a short recurrence at commit time; padded verification shapes support CUDA graphs [7]. Thus read-only speculative verification followed by accepted-path replay is direct prior art, even when the verification arithmetic differs from ours.

SpecLA keeps value tiles of recurrent state on chip across updates within a chain and schedules independent ready chains in parallel. Boundary states connect dependent chains. This directly overlaps both the state reuse and path scheduling in LumoTree. SpecLA buffers accepted factors for a later fused update, whereas our publication stage replays recorded operands through native GDN updates into running rows. Its evaluation uses a pure GDN model; our serving boundary additionally coordinates convolution and softmax attention [6]. SpecLA’s token-replay baseline regenerates factors, so its cost cannot be assigned to our replay of cached operands.

Bole derives a tree-structured closed form and evaluates it with a value-tiled finite-polynomial solver. It reuses intermediate tiles on chip, stores compact update factors, and reconstructs the accepted state. Its SGLang integration combines batch-wide verification budgeting with captured GPU execution. Evaluation includes unquantized Qwen3.5-27B on GB10 and replay of recorded OpenHands agent sessions [28]. This is a direct hybrid-serving comparison. Our evaluation reports recorded-request replay separately from running the coding agent on tasks.

Concretely, Bole’s correction system is $(I + G)U = R$, where G contains strict-ancestor interactions and R is the source correction. Tree depth d gives $G^{d+1} = 0$, so $U = \sum_{m=0}^d (-G)^m R$. This finite identity changes the arithmetic schedule from sequential updates to matrix propagation [28]. Our path preserves sequential updates and replays accepted operands; Section IV-B explains why the numerical comparison remains an empirical question.

TreeWY expresses GDN tree verification as a triangular

TABLE I
RECURRENT VERIFICATION AND CONTINUATION POLICIES. THESE
MECHANISM COMPARISONS DO NOT RANK THE SYSTEMS BY
PERFORMANCE.

Method	Verification work	Continuation state
Weaver [7]	Ancestor-masked triangular solve	Short selected-path recurrence
SpecLA [6]	Resident state tiles; parallel chains	Buffered factors; later fused update
Bole [28]	Finite-Neumann correction solve	Compact factors; state reconstruction
TreeWY [30]	Triangular solve in fused kernel	Prior-path reconstruction with next verify
ReplaySSM [31]	Corrected inputs; cached transitions	Deferred state update
OneLA [32]	Shared-base projections across beams	Compact per-beam transition records
LumoTree	Resident state tiles; parallel paths	Native replay; coordinated convolution and KV publication

system and reconstructs the accepted state from pseudo-values. The author implementation fuses reconstruction of the previous accepted state with the next verification, reusing a resident state tile. Its GDN tree kernel supports CUDA graph capture; the paper’s wider-tree fallback to piecewise execution concerns ancestry masks in the softmax-attention integration [29], [30]. Its B200 evaluation uses greedy verification and disabled prefix caching. A comparison must therefore include fused commit work and distinguish kernel support from whole-model execution policy.

ReplaySSM caches recurrent inputs and delays state materialization. Its GDN path uses corrected inputs from a triangular solve, with commitment represented through buffer progress and later state updates [31]. **OneLA** addresses a related state-sharing problem in recommendation beam search: compact transition records and ancestry indices allow projection computation without reconstructing a full state for each beam, and a fused kernel shares the prompt-derived state across beams [32]. Its beam-search output and commitment policy differ from speculative acceptance. Table I compares state lifetimes; published speed ratios from different models and workloads are not combined into a ranking.

D. Design focus

LumoTree connects path-parallel recurrent computation to one accepted continuation across the hybrid model. The shared descriptor links path boundaries, convolution ancestry, attention visibility, physical KV slots, and the next drafter input. Native replay preserves the serving system’s recurrent-state ownership while selected-path gathering and remapping publish the other state components. Spine-first KV placement controls the physical attention reduction layout within this design, building on established floating-point reduction and batch-invariance concerns [33]. Evaluation separates component behavior from complete agent-workload performance.

III. TREE-VERIFIER ARCHITECTURE

A. State and the execution boundary

Let m be a materialized token prefix: the tokens already consumed by the target’s forward computation. Write its persistent state as

$$\mathcal{S}(m) = (R(m), C(m), K(m)), \quad (1)$$

where R collects recurrent matrices, C collects convolution histories, and K collects attention KV entries. The logical emitted prefix can also include a newly sampled token b that has not yet been consumed. The continuation boundary is then $(\mathcal{S}(m), b)$, not a claim that $\mathcal{S}(mb)$ has already been computed. This distinction is essential when counting the correction or bonus token while checking cache state.

A tree descriptor contains candidate token IDs, parent indices, depth/position information, attention visibility, and maps between logical nodes and physical slots. For a candidate node i , each target layer must consume the state obtained from the candidate’s root-to-parent path. At publication, the selected materialized path, pending-token convention, and next-drafter input row must agree. Figure 2 illustrates the three state surfaces coordinated by the verifier.

B. The recurrence and path locality

For one GDN head, let $R \in \mathbb{R}^{d_v \times d_k}$. Queries and keys are in $\mathbb{R}^{d_k}$ and values in $\mathbb{R}^{d_v}$. With decay α_i and write strength β_i , a node applies the standard update [4]

$$R'_i = \alpha_i R_{\pi(i)}, \quad \delta_i = \beta_i (v_i - R'_i k_i), \quad (2)$$

$$R_i = R'_i + \delta_i k_i^\top, \quad o_i = R_i q_i, \quad (3)$$

where $\pi(i)$ is its parent. A previous node in the packed buffer need not be $\pi(i)$. The causal convolution likewise requires path predecessors rather than adjacent packed rows. The attention visibility relation must select the same logical prefix and ancestry. These requirements follow from the model’s computation; a correct attention mask cannot repair a wrong recurrent parent.

The implementation verifies nodes with branch-local scratch and makes the selected continuation durable through replay and remapping. This is an implementation route, not the only way to satisfy the contract. In particular, rejecting a branch requires that its contents cannot influence future computation; it does not require zeroing every scratch byte if future reads are correctly restricted.

C. A descriptor shared by verification and commitment

Figure 1 shows how a shared descriptor connects drafting to continuation. Native multi-token prediction (MTP) supplies a principal chain and runner-up candidates; suffix prediction extends selected paths within a bounded candidate capacity [34], [35]. Combining retrieval and neural drafting has prior art, including SAM Decoding [36]. The verifier derives attention visibility, physical packing, path scheduling, and the acceptance walk from the same logical topology. Padding permits reusable execution shapes while remaining invisible to attention and selection. The particular tree geometry used in our experiments is specified in Section VI.



Same selected materialized prefix on all three surfaces; correction token b may remain pending for the next forward.

Fig. 2. The accepted-prefix interface. Recurrent, convolution, and attention-KV state must all describe the same materialized prefix. A newly sampled correction or bonus token can remain pending until the next forward.

The attention layout places the principal spine first in the physical suffix and permutes KV write locations and ancestry-mask key columns while retaining logical query-row and position order. Selection remains in logical-node space; accepted-path publication maps those nodes back to their physical slots. This distinction prevents a layout optimization from silently changing which key/value entries a branch reads. Recurrent paths begin at the native pre-tree state or a transient cut state; the device committer publishes the selected recurrent and convolution state into native running rows and re-linearizes its attention KV entries.

Algorithm 1 separates candidate verification, path selection, and state publication. Every publication operation uses the same materialized path A . A newly sampled token b^* can be emitted while remaining pending: its state is created by the next target forward.

IV. GPU SCAN AND ACCEPTED-PATH REPLAY

A. Path-parallel verification and state lifetime

The GDN verifier decomposes the tree into dependent path groups, a scheduling family also studied by SpecLA [6]. A first path starts from the request’s pre-tree state and exports the cut states needed by downstream paths. The second level evaluates independent branch paths from those cut states. Each GPU program covers one path, head, and value tile, retaining the full key dimension in its working state tile. Sequential updates reuse this tile through the path loop rather than loading a full recurrent state for every node. Candidate outputs continue through the model; rounded operands are recorded for later accepted-path replay. The source expresses this local reuse; register allocation and spills remain compiler- and geometry-dependent.

Cut states and operand rings are temporary within a verification step. The durable continuation resides in the request-owned native recurrent and convolution running rows. Replay starts from the pre-tree running state and publishes the accepted frontier back to those rows; prefix-cache snapshots and restores use that native layout. Static work buffers can be reused by CUDA graphs, but they do not replace the request’s persistent state with a branch-leaf pointer.

The two levels execute in separate GPU launches and exchange only the boundary states required by downstream

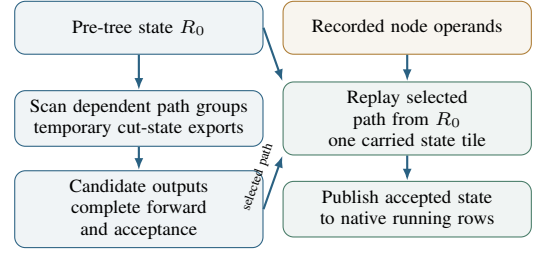


Fig. 3. State lifetime in the path-scan verifier. Temporary cut states connect path groups; recorded operands support selected-path replay. Only the accepted frontier becomes native persistent state.

paths. This schedule exposes parallel branch work while retaining the state-transfer cost at the decomposition boundary.

B. Sequential arithmetic and accepted-path replay

Verification uses sequential path scans with fp32 carried state, raw gate inputs, and in-kernel query/key normalization. Accepted-path replay uses native fused GDN updates with recorded accepted operands and in-place publication to the fp32 state bank. Its graph contains one native update call per GDN layer. Query/key normalization, beta conversion, gate evaluation, and output casts can differ between these implementations; agreement is not guaranteed by their common real-arithmetic recurrence.

The sampling stage produces accepted node indices and path lengths on the GPU. Recurrent replay, convolution publication, and attention-KV publication consume this same path. Graph capture reduces host launch overhead for the sequence of per-layer native updates; each layer retains its own update kernel.

C. Temporary state versus recomputation

Path decomposition caches the states needed where downstream paths branch, then reuses them across independent work. If a program exports N_c cut states with value-tile width B_v and key dimension d_k , their fp32 payload is

$$M_{\text{cut}} = 4N_c B_v d_k \text{ bytes.} \quad (4)$$

This counts one head and value tile’s logical export payload. The current reusable scratch allocation reserves a full matrix slot for each actual node even though only required cut states are written; allocated bytes therefore differ from state traffic. Recomputing ancestry could remove some exports at the cost of repeating updates. Exporting every candidate state would instead trade more writes for less recomputation. The two-level path schedule occupies a particular point in this design space.

Accepted-path replay carries one state tile through a bounded path loop. Its carried-state footprint is $O(B_v d_k)$ in tree size, while the recorded operands and candidate outputs scale with the verification shape. Static padding makes graph buffers and kernel geometry reusable, but padded rows must remain invisible to attention and selection. Neither a shorter logical path nor fewer state writes alone establishes a faster complete request.

The closest scheduling comparison is SpecLA: both systems carry state tiles through sequential paths and parallelize ready

Algorithm 1 Tree verification and native continuation

Input: Native continuation state; candidate tree T ; proposal probabilities q

Output: Emitted tokens; updated native state; next drafter input; any pending token

Verify candidates

- 1: Map T to physical slots, positions, and ancestor masks.
- 2: $p \leftarrow$ target probabilities from path-local GDN, convolution, and attention computation.
- 3: Retain candidate hidden states, rounded replay operands, convolution histories, and temporary KV entries.

Select one continuation

- 4: $(A, b^*) \leftarrow$ accepted materialized path and pending token from sampling with p and q .

Publish the same path across all state components

- 5: **for** each GDN layer **do**
 - 6: Replay recorded operands along A into the native recurrent state.
 - 7: Gather the convolution history for A into the native running row.
 - 8: **end for**
 - 9: Remap the accepted KV entries using the same logical-to-physical slot map.
 - 10: Select the matching hidden-state row for the next drafter input.
 - 11: Advance positions along A ; keep b^* pending until its target forward.
 - 12: **return** Emitted tokens and the updated continuation boundary.
-

paths. Their continuation policies differ. SpecLA buffers accepted factors for application in a later fused update [6]; LumoTree immediately replays recorded operands through native per-layer updates and publishes the matching convolution and attention state. Weaver also uses selected-path replay, but verifies candidates with a triangular solve [7]. Bole reconstructs accepted states from compact factors in a batched commit [28]. TreeWY can fuse prior-frontier reconstruction with the next verification while retaining its working tile [30]. Thus, LumoTree pairs path recurrence with immediate native replay and coordinated hybrid-state publication. This pays explicit accepted-path update work in exchange for returning each iteration to the native state representation. Complete-cycle measurements must include that tradeoff.

V. OPTIMIZING THE LUMOTREE VERIFICATION CYCLE

The architectural choice in Section IV is realized through the implementation changes below. Three representations must agree: logical tree nodes, temporary verification storage, and the native state used by the next forward. LumoTree carries one accepted path across these representations and reorganizes preparation and selection around it, using established techniques such as fusion and graph replay. The nontrivial constraints arise at their boundaries: changing a slot order, a rounding step, or a tied candidate can change which continuation the next iteration consumes. Table II summarizes the changes and these constraints.

A. One device-selected path drives native continuation

Selecting valid output tokens does not by itself update a hybrid model’s state. The recurrent matrix must contain the accepted updates, convolution must retain the accepted history, attention must reference the accepted KV entries, and the drafter must receive the corresponding hidden row. Reconstructing these choices separately on the host introduces

path-list preparation and synchronization between acceptance and publication.

LumoTree replaces that handoff with fixed-shape device products: accepted node indices, path lengths, emitted tokens, and the selected final row. These products feed native recurrent replay, convolution gathering, KV remapping, and drafter-row selection. Recorded operands are staged for the native per-layer updates in one captured committer graph; convolution and KV publication use the same selected path. Graph capture amortizes launch orchestration, while the shared path prevents each state component from independently interpreting acceptance.

The distinction between emitted and materialized tokens must survive this handoff. A correction or bonus token may be emitted while remaining pending for the next target forward; padded path entries must not advance state. After publication, the implementation stores persistent state in native request rows and branch cut states in temporary scratch. The native representation supports the intended prefix-cache and serving-graph interface. The implementation captures the serving, drafter, and committer stages separately.

B. One slot map governs attention and publication

An ancestry mask specifies visible keys, but does not specify their physical placement or the reduction groups that process them. Interleaving masked branch columns with spine keys can change the online-softmax reduction layout even when logical visibility is unchanged. Physical ordering therefore matters both for addressing and for finite-precision execution [33].

LumoTree places the principal spine contiguously in the candidate suffix while retaining logical query-row and position order. It applies the same permutation to KV write slots and ancestry-mask key columns. Acceptance remains in logical-node space; publication translates the selected nodes through that slot map into native continuation positions. Reordering only writes, only masks, or only the final copy would leave another

Algorithm 2 Path-parallel scan and accepted-path replay

Input: Candidate tree T and the pre-tree recurrent state for each GDN layer

Output: Recurrent outputs for all candidates; native recurrent state for the accepted path

Verify paths; keep branch state temporary

```
1: for each GDN layer reached during the target forward do
2:   Compute and record this layer's rounded node operands.
3:   for each path group, in dependency order do
4:     parallel for each ready path, head, and value tile do
5:       Load the working state tile from the pre-tree state or a saved parent cut state.
6:       for each node on this path, in ancestry order do
7:         Update the working tile with this node's operands; write its recurrent output.
8:         Save the tile if a downstream path needs this node's state.
9:       end for
10:    end for
11:  end for
12: end for
```

Finish verification and select a path

13: Complete the target forward and sampling to obtain the accepted materialized path A .

Replay only the accepted path into native recurrent state

```
14: for each GDN layer do
15:   Start from this layer's pre-tree recurrent state.
16:   for each node on  $A$ , in path order do
17:     Apply the native GDN update using this layer's recorded operands.
18:   end for
19:   Publish the resulting recurrent state to the request's native running row.
20: end for
```

TABLE II

CONCRETE IMPLEMENTATION CHANGES AROUND THE PATH VERIFIER. THE THIRD COLUMN IDENTIFIES THE COORDINATION OR NUMERICAL CONSTRAINT THAT MAKES EACH CHANGE NONTRIVIAL. WORKLOAD RATES MEASURE THEIR COMBINED EXECUTION.

Starting work or constraint	LumoTree change	What must remain consistent
Host path-list construction between acceptance and publication	Device-selected path feeds captured native replay, convolution, KV and drafter selection	All consumers select the same materialized prefix; pending and padded tokens do not advance state
Logical ancestry does not determine physical KV placement	Spine-contiguous slots with coordinated writes, mask columns and accepted-path remap	Logical query order, physical source slots and native destination positions agree
Per-node convolution histories and repeated layer preparation	Static ancestry gathers, fused ordered taps and per-layer operand buffers	Tap order, cast boundaries, history padding and layer identity
Separate score preparation and argmax/top-k operations	One shared logits tensor and fused selection into graph buffers	Both reference tie conventions, candidate order and duplicate behavior
Repeated KV staging and limited decode parallelism	Grouped query heads and split-K in the patched tree-attention kernel	Tree visibility and slot mapping across partitions; characterized reduction error

consumer reading a different branch. The implementation change is this coordinated mapping across verification and publication.

The patched FlashAttention-2 decode kernel combines this layout with grouped query heads that share staged KV tiles and split-K context partitions. KV reuse and partitioned attention are established techniques [21], [22]; here they are applied under the tree's visibility and slot mapping. They target repeated KV staging and insufficient parallel work at batch one. Prefill retains a native-compatible attention path. The current component tests measure repeatability and numerical error of the deployed attention kernel; the isolated effect of spine placement remains

an ablation question.

C. Precompute branch histories while preserving arithmetic

The starting tree-convolution implementation constructs candidate histories with per-node operations and repeats path-row preparation across layers. LumoTree derives static gather indices from the topology, gathers windows from the prior convolution history and candidate inputs, and fuses the tap computation. Stacked operand buffers retain each GDN layer's inputs for subsequent accepted-path replay.

This transformation must preserve more than ancestry. Convolution tap order and cast boundaries affect the values

recorded for recurrent updates, and unused history positions must retain the intended padding. A generic reduction can compute the same real-arithmetic sum in a different floating-point order. The fused implementation therefore retains the ordered tap operations and conversion points while replacing repeated history construction with prepared data movement. Each layer still records and replays its own operands.

D. Fuse draft selection without changing proposals

At each MTP head depth, LumoTree computes one full-vocabulary logits tensor and reuses it for the principal token and sibling candidates. A fused CUDA kernel writes the spine token and three ordered candidate indices directly into reusable graph buffers, replacing separate argmax/top-k reductions and their output copies. The principal chain advances through captured MTP iterations; sibling selection reuses its scores, and suffix prediction extends selected paths within the fixed verification capacity.

The subtlety is that argmax and top-k do not have interchangeable tie behavior. The reference argmax chooses the lowest maximal token index, while its top-k operation orders selected ties differently. Simply taking the first top-k result as the spine token can change the proposal tree. Our fused selection preserves both reference outputs, including duplicate proposals where produced by the reference. Exact-output and captured-replay tests check this component. The optimization removes redundant score preparation and selection work while retaining the vocabulary projection.

VI. EVALUATION SETUP

We evaluate Qwen3.8-27B NVFP4 on a single NVIDIA DGX Spark (GB10) through two protocols. In the coding-agent deployment, Qwen Code performs repository edits and tool calls on ten SWE-bench Verified Astropy tasks [8], with one attempt per task; the benchmark evaluator tests the resulting patches. A separate replay compares serving configurations on 43 recorded requests. Decode rates pool token counts and decode durations over the same completed requests, excluding time to first output. Agent duration additionally includes tools and repository operations; server startup and evaluation are measured separately.

The evaluated tree has 27 draft candidates and one root, padded to 32 verification rows. A five-depth MTP head supplies 15 candidates; suffix prediction adds a six-token principal tail and rescue branches of four and two tokens. Maximum draft depth is eleven, and four post-root MTP forwards construct the head. Attention groups query heads in pairs and divides the context into four partitions. The model has 48 GDN layers and sixteen full-attention layers. All LumoTree workload deployments use this geometry and the patched FlashAttention-2 path. Appendix A gives the serving configuration; the accompanying artifact identifies each source and kernel version.

The replay corpus contains 43 requests recorded from the Astropy tasks 12907 and 13033. Each configuration receives the same saved messages and tool definitions, sequentially with at most one active request. Reported vLLM prompt lengths

range from 23,024 to 68,557 tokens. Sampling uses temperature 0.6, top- p 0.95, top- k 20, and presence penalty 1.0, with a 1,024-token response cap and no per-request seed. This protocol measures fixed-request serving behavior; it does not rerun either coding task.

Native autoregressive decoding, chain MTP, and LumoTree use the same checkpoint, vLLM image, tokenizer, chat template, and common serving settings, including prefix caching, chunked prefill, a 131,072-token context limit, and one active sequence. The native arms use their native attention invocation; LumoTree uses its specialized tree-attention path. SGLang EAGLE uses the same checkpoint and saved request files, with FlashInfer attention and its own serving and tool-parsing implementation. Although both stacks receive the same chat-template file, SGLang reports 146 additional prompt tokens per request. Whether this difference comes from serialization or token accounting remains unresolved.

The replay client measures time from the first nonempty streamed content, reasoning, or tool-call delta to stream completion. We apply Eq. 5 to these client durations and server-reported completion counts, pooling matched token and time totals across requests and repetitions. The estimate excludes first-output latency and server startup. Subsequent stream delivery and parser behavior remain in the measured duration.

VII. RESULTS

A. SWE-bench Verified coding-agent workload

We deploy LumoTree with Qwen Code on ten Astropy tasks from SWE-bench Verified. The agent inspects repositories, edits files, executes tools, and produces a patch for each task. Model serving runs on a single GB10. Six patches resolve their tasks and four fail tests; agent durations range from approximately 2 to 41 minutes. Appendix B gives the task-level outcomes.

For completed request r , let n_r be the output-token count, ℓ_r the server-recorded end-to-end latency, and f_r the time to first output. We report the pooled decode rate

$$D_{\text{pool}} = \frac{\sum_{r=1}^R (n_r - 1)}{\sum_{r=1}^R (\ell_r - f_r)}. \quad (5)$$

Counts and durations cover the same completed requests, including requests that summarize the agent’s context when it grows too long. The numerator counts output intervals, and the denominator excludes first-output latency and between-request agent/tool work. Thus, D_{pool} measures token production over accumulated request durations rather than aggregate wall-clock throughput.

The ten-task deployment records 25.63 pooled tokens/s over 265 completed requests. Appendix A gives the serving configuration. This descriptive rate has no matched native-decoding or chain-MTP baseline (Section IX).

B. Recorded-request replay

Table III reports eighteen complete configuration runs, each containing all 43 requests. Across three runs per arm, LumoTree records 28.79 pooled tokens/s and five-token native

TABLE III

SAMPLED REPLAY OF 43 RECORDED REQUESTS, CAPPED AT 1,024 OUTPUT TOKENS. RATES POOL MATCHED TOKEN AND TIME TOTALS ACROSS ALL RUNS. RANGES DESCRIBE INDIVIDUAL RUN RATES, NOT CONFIDENCE INTERVALS. SGLANG ROWS USE A DIFFERENT STACK WITH UNRESOLVED PROMPT-TOKEN ACCOUNTING.

Configuration	Runs	Tokens/s	Run range
vLLM autoregressive	1	10.54	—
vLLM MTP, 1 token	1	15.37	—
vLLM MTP, 3 tokens	1	22.06	—
vLLM MTP, 5 tokens	3	26.12	26.04–26.23
vLLM MTP, 7 tokens	1	25.72	—
LumoTree	3	28.79	28.38–29.41
SGLang EAGLE, 3/4	1	26.32	—
SGLang EAGLE, 5/6	3	29.28	29.23–29.36
SGLang EAGLE, 7/8	3	30.05	29.18–30.69
SGLang EAGLE, 9/10	1	28.81	—

TABLE IV

EXPLORATORY REPLAY VARIANTS, ONE COMPLETE RUN EACH. THESE MEASUREMENTS DO NOT ESTABLISH ISOLATED OPTIMIZATION EFFECTS.

Disabled optimization	Tokens/s
Grouped split-K tree attention	27.11
Fused draft top- k selection	29.40
Precomputed tap-operand preparation	29.02

MTP records 26.12, a 10.2% difference on this corpus. Their individual run rates range from 28.38 to 29.41 and from 26.04 to 26.23 tokens/s, respectively. Plain autoregressive decoding records 10.54 tokens/s in one run. The native MTP sweep uses one, three, five, or seven draft tokens; five-token MTP has the highest observed pooled rate among these settings.

For SGLang, the paired setting denotes speculative steps and draft-token capacity; EAGLE top- k is one throughout. Seven steps and eight draft tokens yield 30.05 pooled tokens/s over three runs. Differences in prompt-token counts, attention, prefill settings, and tool parsing limit this cross-stack comparison (Section IX). The sweep is exploratory and has unequal repetition counts.

Three additional runs disable individual LumoTree optimizations (Table IV). Each variant completes all 43 requests once. The stock-attention variant records 27.11 tokens/s, while disabling fused draft selection or precomputed tap-operand preparation records 29.40 and 29.02. Stochastic output lengths and limited replication prevent attributing these differences to the individual transformations. The variants do not yet provide a per-optimization latency or memory decomposition.

Output-side counters pooled over the repeated vLLM runs report 4.47 accepted drafts per speculative event for LumoTree and 3.17 for five-token MTP. These counters describe emitted drafts, excluding the correction or bonus token, and do not identify the committed path. SGLang reports only windowed acceptance averages, so no corresponding cumulative acceptance comparison is reported.

TABLE V

ACCEPTANCE AND MEAN GPU SPAN DURATIONS FOR THE TEN-TASK DEPLOYMENT. DISPATCH INCLUDES ACCEPTANCE, PATH HANDLING, AND STATE COMMITMENT.

Metric	Value
Accepted drafts per event	4.05
Target span (ms)	114.89
Drafter span (ms)	54.76
Dispatch span (ms)	20.15

TABLE VI

GDN COMPONENT CHECKS ON GB10. BITWISE COLUMNS COMPARE WITH THE NATIVE CHAIN; THE PAIRED RULE COMPARES BOTH IMPLEMENTATIONS WITH A FLOAT64 REFERENCE. COUNTS ARE PER BLOCK AND PROCESS.

Operand sets	BF16 output bitwise	FP32 state bitwise	Paired rule passed
Calibration	3,004/3,072	2,688/2,688	5,760/5,760
Held-out	3,018/3,072	2,688/2,688	5,760/5,760

C. Acceptance and execution costs

The ten-task deployment emits an average of 4.05 accepted drafts per speculative event. This counter measures output draft length after filtering, excluding the correction or bonus token. It does not identify the branch published into persistent state. Physical draft slots also include padding and are distinct from logical candidates.

Table V summarizes the phase measurements. Each timer is reduced using its own duration sum and span count, with no outstanding samples at task boundaries. Dispatch combines sampling and native replay; the spans do not form an isolated critical-path decomposition.

D. Numerical and candidate-selection validation

We test GDN verification and accepted recurrent-state publication on two calibration and two held-out synthetic operand sets. Each set supplies 48 recurrent instances with the evaluated tree geometry: 32 physical output rows, including padding, and 28 accepted publication paths. Each block runs twice in each of two fresh processes. These repetitions check reproducibility; they do not increase the number of distinct inputs.

For each head, we compare both implementations with a float64 recurrence. We require both root-mean-square and maximum absolute errors to satisfy

$$E_{\text{tree}} \leq 1.10E_{\text{native}} + 2^{-24}M_{64} + 2^{-149}, \quad (6)$$

where M_{64} is the corresponding root-mean-square or maximum magnitude of the reference. Table VI reports all enumerated cases passing in both processes. All published recurrent states are bitwise equal to the native chain, while 68 calibration and 54 held-out output tensors differ in BF16 rounding. Native and candidate results reproduce bitwise across repeats and processes.

Calibration controls detect sibling substitution, swapped state rings, and stale publication metadata. Poisoning an off-path

TABLE VII

RECURRENT AGREEMENT ON GB10 UNDER DISTINCT SEQUENTIAL REFERENCES. COUNTS GIVE CELLS OUTSIDE THE PAIRED ERROR BOUNDS AMONG 64,512 OUTPUT CELLS AND 6,912 STATE CELLS PER CONFIGURATION, WITH IDENTICAL COUNTS IN BOTH REPEATS. THE TWO REFERENCE GROUPS ARE NOT A COMMON ACCURACY RANKING.

Configuration	Output cells	State cells
<i>Native GPU recurrence reference</i>		
LumoTree	0	0
<i>Sequential software references</i>		
Weaver author-default	53,745	1,154
Weaver (FP32 preparation)	57,707	388
TreeWY author-default	37,528	6,912

branch leaves the tested selected-path state unchanged, and a zero-replay control is rejected. These results support the tested GDN scan and recurrent publication mechanism.

The fused full-vocabulary selection test compares the two output tensors against the reference argmax and top-k operations. Across 1,368 input cases and five block-count settings (6,840 configurations), it records zero byte mismatches; planted negative controls fire. A separate captured four-level selection graph completes 24 replays with zero mismatches. The reference selection outputs match exactly on these inputs and graph replays.

Attention tests use the same kernel build as the ten-task deployment. They show bit-identical repeated output and log-sum-exp values within each of sixteen determinism cases, with matching digests across two processes. On synthetic query/key/value tensors drawn at scales measured from model operands, 93.31% of output elements agree with the reference kernel within two ULP, the maximum absolute output difference is 2^{-8} , and maximum log-sum-exp disagreement is four ULP. All nine checks pass, including comparisons against a float64 dense reference and checks for non-finite disagreements. Split-K changes the reduction order, so its bounded error accompanies the repeatability result.

Runtime traces confirm execution of the specialized kernel in all sixteen attention layers, fused selection, device acceptance, and captured state publication during the workload runs.

E. Recurrent agreement under distinct references

We compare recurrent verification and accepted-state publication using held-out synthetic operands spanning 48 recurrent instances and 48 value heads. Verification covers the tree’s 28 active nodes. Three accepted-path publications produce a history of 1, 6, and 11 updates, including Weaver’s accepted replay and TreeWY’s final deferred commit. Each configuration is evaluated twice in one process, with a fresh initial state for each cycle. A numerical cell fixes the recurrent instance, output node or publication, and value head; errors are reduced over the remaining vector or matrix entries.

Each cell must satisfy both root-mean-square and maximum absolute error bounds,

$$E_{\text{method}} \leq 1.10E_{\text{seq}} + 2^{-24}M_{64} + 2^{-149}, \quad (7)$$

where errors are measured against the specified float64 recurrence and M_{64} is its corresponding root-mean-square or maximum magnitude. For LumoTree, the sequential comparator is the native GPU recurrence. Weaver and TreeWY use sequential software comparators that do not emulate every tensor-core rounding operation.

TreeWY retains the author implementation’s normalization floor of 10^{-12} and BF16 dot products. Weaver uses TF32 verification, FP32 elementwise accepted replay, and an additive normalization constant of 10^{-6} . We report both its author-default preparation and a variant with FP32 preparation. The two Weaver variants retain their respective preprocessing conventions.

Table VII reports the cells outside each implementation’s paired bounds. All configurations produce finite tensors and reproduce bitwise between repeats. TreeWY’s BF16 compact commit and its sequential comparator’s FP32 outer-product updates, for example, are different floating-point computations. The normalization and arithmetic conventions are part of the numerical comparison and must be interpreted alongside its reference choice (Section IX).

VIII. DISCUSSION

The design couples tree geometry, physical layout, and continuation publication. Optimizing any one of these in isolation can move work elsewhere or invalidate another stage’s mapping. Transient cut states, graph buffers, and native persistent rows serve different purposes: their separation makes the storage and recomputation tradeoff explicit. Path scans retain sequential arithmetic, while native replay pays accepted-path update work to return persistent state to the serving representation.

For coding agents, decoding rate and task completion measure different effects. Additional tokens can reflect more reasoning or a longer tool loop, and tool execution contributes to agent duration without entering the decode denominator. Recorded-request replay fixes the incoming messages but does not execute the newly generated tools or subsequent agent decisions. Its 10.2% rate advantage over five-token native MTP therefore applies to this request corpus and serving configuration. Task outcomes and GPU spans complement the pooled rate: they describe application behavior and the work performed by the serving cycle.

IX. LIMITATIONS

a) *Numerical scope:* The component results cover recurrent verification and publication, attention error, and candidate selection. Full-model tree equivalence remains untested: across processes, native target/draft choices agree on 68 of 84 paths despite stable greedy decisions (Appendix C). Multi-cycle continuation, prefix-cache cold/hit behavior, row reuse, stale-state refusal, and distribution preservation remain open. The recurrent results for LumoTree, TreeWY, and Weaver use different sequential comparators and preprocessing conventions; they measure agreement with those references, not a common accuracy or timing ranking. Matching starting states, tokens,

and positions also does not guarantee identical intermediate hidden tensors, so the observed native differences do not identify a kernel-level cause.

b) Performance and task scope: The ten-task deployment contains one attempt per selected task and no matched native-decoding or chain-MTP control. Its task outcomes provide no benchmark-wide quality estimate or controlled task-completion speedup. The separate replay adds native controls and three repetitions for LumoTree and five-token MTP, but covers requests from only two agent sessions. Sampling produces different output lengths, and the response cap truncates some requests. The timing estimate also includes stream buffering and parsing: subtracting one token approximates the first output interval when a speculative chunk contains several tokens. SGLang’s reported prompt counts differ from vLLM’s by 146 tokens on every request; identical tokenized inputs have not been established. The configuration sweep and single-run variants do not establish significance or isolate optimization effects. Equal memory-fraction settings do not establish equal KV-block capacity. GPU instrumentation overhead has not been measured, and logical state-storage formulas are not measurements of allocated scratch memory.

X. CONCLUSION

LumoTree combines path-parallel recurrent scans with immediate native replay and a shared mapping for convolution, attention, acceptance, and drafter continuation. The implementation integrates fused selection, device-resident acceptance, and grouped split-K attention. Component evaluation characterizes exact selection outputs, bounded recurrent error, and attention repeatability. On 43 recorded requests, sampled replay on GB10 yields 28.79 pooled tokens/s versus 26.12 for five-token native MTP on the same vLLM stack. A separate ten-task coding-agent deployment records 25.63 pooled tokens/s. These results support a bounded serving-rate comparison; full-model continuation and controlled task-completion improvements require further evaluation.

APPENDIX A

SERVING CONFIGURATION AND MEASUREMENT

The accompanying artifact [37] provides the implementation, launch settings, task metadata, and raw measurements for the coding-agent deployment and component studies, including failed, excluded, and incomplete attempts. The recorded-request replay has a separate source and raw-data supplement; its input manifest, run selection, and pooled-rate audit are preserved with this manuscript source.

Workload rates use Eq. 5; token counts and latency sums cover the same completed requests. Metric counters are checked for continuity, resets, and outstanding requests at task boundaries. First-output latency, tools, and evaluator time remain distinct from decode duration. GPU phase times divide each timer’s duration sum by its own span count. Output-side acceptance counters describe emitted draft lengths and do not identify the committed tree path.

TABLE VIII
SERVING CONFIGURATION FOR THE TEN-TASK LUMOTREE DEPLOYMENT. EXACT SOFTWARE VERSIONS AND LAUNCH SETTINGS ACCOMPANY THE ARTIFACT.

Setting	LumoTree / vLLM
Target	Qwen3.8-27B; ModelOpt mixed NVFP4, BF16 execution
Proposals	27 drafts plus root, padded to 32 rows; maximum depth 11
Attention	Patched FlashAttention-2; head grouping 2, split-K 4
Context	131,072 tokens
Scheduling	4,096 scheduled tokens; one sequence
Response cap	24,000 tokens
Sampling	Temperature 0.6; top- p 0.95; top- k 20; min- p 0; presence penalty 1.0
Agent	Qwen Code 0.19.4; one task at a time

For target probabilities p and proposal probabilities q , ordinary rejection sampling combines accepted mass $\min(p, q)$ with residual mass proportional to $(p - q)_+$. This gives the target marginal for the probabilities actually supplied [1], [2]. Serving additionally requires that subsequent forwards consume the state associated with the accepted path.

APPENDIX B

ADDITIONAL WORKLOAD RESULTS

Table IX reports the ten-task deployment, with one attempt per task. All attempts use tools and produce nonempty patches. Agent time includes model service and tool execution, excluding server startup and subsequent evaluation.

TABLE IX
TASK-LEVEL RESULTS FOR THE TEN-TASK LUMOTREE DEPLOYMENT. TASK IDS HAVE PREFIX ASTROPY___ASTROPY-. DURATIONS ARE ROUNDED TO MINUTES.

Task suffix	Agent time (min)	Evaluation
13977	25	Failed tests
14096	23	Resolved
14182	7	Failed tests
14309	2	Resolved
14365	7	Failed tests
14369	28	Resolved
14508	41	Resolved
14539	19	Resolved
14598	26	Failed tests
14995	5	Resolved

APPENDIX C

NATIVE-REFERENCE REPRODUCIBILITY

We examine native-reference reproducibility across independent processes as a prerequisite for full-model tree validation. Each condition covers 84 paths: a root-only continuation and 27 candidate paths for each of three calibration prefixes. Every path is repeated twice in each of two independent processes. The conditions distinguish natural prefill from an imported common starting state and separate target continuation from continuation coupled to the MTP drafter.

TABLE X

CROSS-PROCESS NATIVE-REFERENCE AGREEMENT, IN PATHS OUT OF 84. STATE/LOGITS REQUIRES IDENTICAL PUBLISHED TARGET STATE AND FULL TARGET LOGITS. JOINT CHOICES REQUIRES THE NEXT TARGET DECISION AND THE ACTUAL MTP SPINE AND ORDERED TOP-THREE SELECTIONS AT EVERY TESTED PHASE. A DASH DENOTES AN UNTESTED JOINT MTP CONDITION.

Starting-state control	Greedy	State/ logits	Joint choices
Natural prefill	70/84	0/84	–
Common target state	84/84	84/84	–
Common target and MTP state	84/84	65/84	68/84
With fixed kernel selection	84/84	65/84	68/84

Natural-prefill continuation reproduces within each process but changes the greedy winner on fourteen paths across processes. The target-only control with common starting state reproduces identical target state, logits, and greedy decisions on all paths. When MTP continuation is included, greedy decisions remain stable on all paths, but target state and full logits agree on only 65, and joint target/draft choices on 68. Holding kernel selection fixed yields the same aggregate counts.

All sixteen paths with differing joint choices have long prefixes. Their within-process repetitions agree, and every principal MTP spine remains stable; the disagreements affect secondary top-three membership or order. The tested phases include initial MTP continuation, a follow-up, and continuation after consuming the pending target token. Common imported state and matched tokens and positions do not guarantee identical intermediate hidden tensors.

The sixteen joint-choice disagreements prevent this reference from supporting a full-model tree equivalence claim (Section IX).

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